



Apatite

△ Fine, medium-blue, step-cut oval apatite

The name **apatite** is derived from the Greek *apate*, meaning "deceit", as it often resembles the crystals of other minerals such as aquamarine, amethyst, and peridot. It can be intensely coloured, occurring in vivid greens, blues, violet-blues, purples, and rose-reds. As a relatively soft crystal, it is not widely used as a gemstone, and although transparent apatite is sometimes faceted and mounted in jewellery, care must be taken when wearing it as it can scratch. Some of the largest apatite crystals have been found in Canada, weighing up to 200kg (485lb).

Specification

Chemical name Fluorapatite, chlorapatite, hydroxyapatite
Formula $\text{Ca}_5(\text{PO}_4)_3(\text{F}, \text{OH}, \text{Cl})$ | **Colours** Various | **Structure** Hexagonal or monoclinic | **Hardness** 5 | **SG** 3.1–3.2
RI 1.63–1.64 | **Lustre** Vitreous, waxy | **Streak** White
Locations Madagascar, Brazil, Myanmar, Mexico



Apatite and muscovite | Rough | Fine green apatite crystals can be seen growing within the white tabular muscovite, or mica, crystals in this rough specimen.

Muscovite or common mica crystals



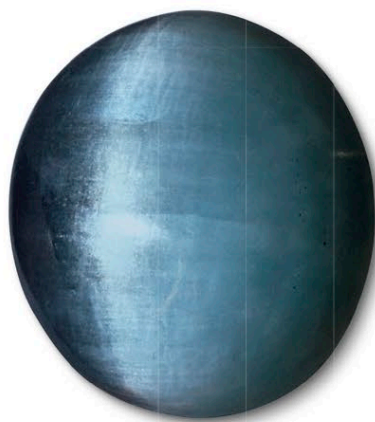
Apatite in calcite | Rough | Apatite is found in a number of geological environments, seen here as green crystals within a calcite groundmass.



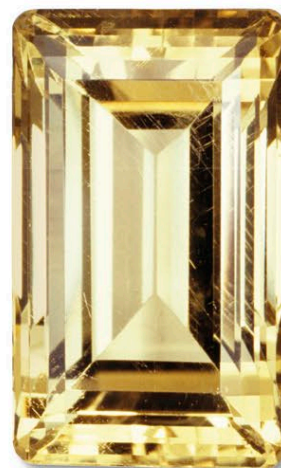
Prismatic crystal | Rough | Apatite can look deceptively similar to other crystals. To an amateur, the blue crystals here could be mistaken for aquamarine.



Mexican apatite | Rough | Yellow apatite crystals, such as this Mexican example with a pyramidal end and a hexagonal prism, is popular with collectors and jewellery-makers.



Cabochon of apatite | Cut | An attractive dark blue specimen, this apatite has been cut and polished *en cabochon* and shows a cat's eye effect.



Step-cut apatite | Cut | Yellow apatite crystals from Durango, Mexico, were often faceted into fine gems, such as this rectangular step-cut stone.



Oval brilliant | Colour variety | Apatite gems displaying a fine blue colour, such as this 6.16-carat blue oval brilliant, are among the most popular varieties.